

While Loop Diagnostic test

Name (caps) _____ Student ID (9 digits)

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Lab Time _____ Desk Number: Desk-_____ (eg Desk-G02-060)

How to work out what a code fragment does: draw a table

The simplest and most reliable way to work out what a code fragment does (and to understand) is to draw a table. This is called tracing or dry running code, Let's work out what this fragment does:

```
int total = 0;           // L1
while (total < 10)       // L2
{
    System.out.println(total); // L3
    total = total + 5;       // L4
}
System.out.println("END"); // L5
```

Executing Line L1

Line L1 initialises total, so total is set to 0.

LINE	total	total<10?	Printed
L1	0		

Executing Line L2

Line L2 is the test of the loop. The boolean expression evaluates to true or false and only enters the loop if it is true. Here it tests if total < 10 ie 0 < 10. This is true so it enters the loop.

LINE	total	total<10?	Printed
L1	0		
L2		0<10? true	

Executing Line L3 and then Line L4

This prints total, which is 0, moving the cursor to the next line. Line L4 adds 5 to total making it now 5.

LINE	total	total<10?	Printed
L1	0		
L2		0<10? true	
L3			0
L4	5		

Executing Line L2

We return back to line L2 and the test is checked again. It is still true as 5 < 10 so we enter the loop again.

LINE	total	total<10?	Printed
L1	0		
L2		0<10? true	
L3			0
L4	5		
L2		5<10? true	

Continued overleaf

Executing Line L3 and then Line L4

We again print total which is 5 then update it by adding 5 to it to make it 10.

LINE	total	total<10?	Printed
L1	0		
L2		0<10? true	
L3			0
L4	5		
L2		5<10? true	
L3			5
L4	10		

Executing Line L2

We return back to line L2 and the test is checked again. It is now false as 10<10 is false so we exit the loop jumping beyond the loop.

LINE	total	total<10?	Printed
L1	0		
L2		0<10? true	
L3			0
L4	5		
L2		5<10? true	
L3			5
L4	10		
L2		10<10? false	

Executing Line L5

The next line after the loop is L5. It just prints END

LINE	total	total<10?	Printed
L1	0		
L2		0<10? true	
L3			0
L4	5		
L2		5<10? true	
L3			5
L4	10		
L2		10<10? false	
L5			END

Overall what is printed is:

0
5
END

You must show all your working in the following. No correct table (exactly the same as the solution)... no marks.

1. What is printed when the following code fragment has executed?

```
int t = 3; // L1
while (t <= 7) // L2
{
    t = t + 2; // L3
    System.out.println(t); // L4
}
System.out.println(t); // L5
```

Printed:

LINE	t	t<=7?	Printed

2. What is printed when the following code fragment has executed?

```
int t = 8; // L1
while (t <= 7) // L2
{
    t = t + 2; // L3
    System.out.println(t); // L4
}
System.out.println(t); // L5
```

Printed:

LINE	t	t<=7?	Printed

3. What is printed when the following code fragment has executed?

```
int t = 7; // L1
while (t <= 7) // L2
{
    System.out.print(t); // L3
    t = t + 2; // L4
}
System.out.println(t); // L5
```

Printed:

LINE	t	t<=7?	Printed

4. What is printed when the following code fragment has executed?

```
int x = 2; // L1
while (x < 8) // L2
{
    x = x + 3; // L3
}
System.out.println(x); // L4
```

Printed:

LINE	x	x<8?	Printed